ERHVERVSAKADEMI KØBENHAVN

MoSec Part II — ACL Security Assessment

Mandatory Deliverable II — Network & Communication Security



Objective: Document and evaluate your group's ACL security approach for MoSec Ltd.

Your report must demonstrate how your implemented ACL rules satisfy the organization's business security requirements, supported by ICMP testing and packet evidence.

Important: The objective is not simply to demonstrate working ping commands. You must explain how your ACL strategy, rule ordering and observed traffic collectively demonstrate that the business security requirements have been fulfilled.

1. Business Security Requirements

ID	Source	Destination	Action
BR01	Admin	Employee, Server, Printer, Guest, Internet	ALLOW
BR02	Employee	Admin, Server, Printer, Internet	ALLOW
BR03	Guest	Admin LAN, Employee LAN, Server LAN, Printer LAN	DENY
BR04	Guest	Internet	ALLOW

You must determine the required IP addresses, CIDR ranges, ACL conditions and rule priorities themselves. You may combine requirements into fewer ACL rules if you can justify your approach.

2. Spoofing and ACL Rule Ordering

Your security assessment must also address:

- **IP spoofing:** Demonstrate the Guest-origin spoofing scenario, using Admin's IP address as the forged source.
- **Anti-spoofing:** Design and implement an ingress ACL mitigation, then verify its effectiveness.
- **Rule priority:** Demonstrate how incorrect ACL rule ordering can affect security and how changing priorities resolves the issue.
- **Additional spoofing scenarios:** Investigate other internal addresses Guest could impersonate, refine your ACL configuration if necessary, and verify Guest Internet access remains functional.

3. Mandatory Report Contents

Your report must include:

1. **Security strategy:** Explain your overall approach to satisfying the business requirements.
2. **Final ordered ACL table:** Show all implemented rules, including priority, source, destination, protocol, action and relevant interface where applicable.
3. **ICMP verification:** Document the expected and actual result of each required communication flow, identifying the relevant ACL rule.
4. **Spoofing analysis:** Explain the observed traffic before and after implementing anti-spoofing, including rule priority.
5. **Conclusion:** Evaluate whether your final ACL configuration satisfies the organization's security requirements.

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4. Mandatory Evidence and Appendix

- Include screenshots showing successful and unsuccessful ICMP ping tests.
- Each screenshot must identify the source, destination, expected result and observed result, and reference the relevant security requirement.
- For denied connections, a ping timeout alone is insufficient to conclusively attribute the result to an ACL. Support your findings with firewall rule counters or packet captures.
- For spoofing, include before-and-after evidence demonstrating whether forged packets reached their destination and whether the anti-spoofing rule matched them.

5. Submission Requirements

Submit one PDF document — maximum 5 physical pages. The title page, table of content & appendix are not counted! *There is no word-count requirement.*